

Short communication

Instantaneous frequency estimation based on the
linear canonical transform[☆]

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Abstract

The relationship between the linear canonical transform power spectrum and signal's phase differentiation operation has been investigated. Based on this relationship, a novel method for solving the instantaneous frequency estimation problem is proposed. The advantage of this method is demonstrated by two examples, namely, signals with and without noise. The proposed method is non-iterative, and it does not need projection or peak-searching.

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1. Introduction

The instantaneous frequency (IF) of a signal is a parameter which is of significant importance in practice. In many situations such as seismic, radar, and sonar, communications, and biomedical applications, the IF is a good descriptor [1,2]. Especially in the signal processing and communications, estimating the IF of non-stationary signal is of great importance. Boashash [1,2] systematically describes the definition of IF and reviews the IF estimation methods, for example, central finite difference [3], time–frequency distribution [4,5] and so on. However, the non-time–frequency methods are sensitive to the noise, so its application is largely restricted [3]. Although time–frequency methods can give the time-varying spectrum of a signal in the time–frequency plane, which largely improved the accuracy of IF estimation, it takes high computing cost and computational complexity [4,5]. Moreover,

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for the existence of time–frequency coupling of signal and noise, that is, the projection of the signal and noise has overlapping in the time–frequency plane, it cannot filter out the noise [6,7]. Cong et al. [8,9] use the recursive algorithm and the square of the signal modulus to estimate the IF, which causes large computation and complexity, so it is very necessary to look for non-recursive methods to estimate the IF.

In recent years, for its unique advantage, the linear canonical transform (LCT) is shown to be one of the important tools in signal processing community [10–12]. For example, we can obtain good results of filtering and interference separation in the suitable LCT domain as long as the signal and noise are linearly separable [13]. Based on the existing results of the LCT, this paper shows a novel method for solving the IF estimation problem from the LCT power spectrum and signal's phase differentiation, which does not need iterations to achieve the results with high precision yet at a low cost of computation.

This paper is organized as follows: Section 2 introduces some preliminary results associated with the IF and the LCT. Section 3 discusses the method of instantaneous frequency estimation based on the LCT. The simulation results are proposed in Section 4. Section 5 concludes the paper.

2. The preliminary

2.1. The instantaneous frequency (IF)

The IF is a fundamental concept not only in communication engineering, but also in nature. In order to give an exact definition of IF, many people have made a big try. Carson and Fry in 1937 [14] first considered a variable frequency in an electric theory context and then applied the concept to frequency modulated signal. Then, Van der Pol in 1946 [15] arrived at a similar definition of IF to that of Carson and Fry by analyzing an expression for simple harmonic motion. Moreover, several other scientists [16,17] have also contributed to the study of IF. Many definitions have been given, but it is commonly accepted that the IF is the derivative of the phase of the signal [1]. Mathematically, the IF is defined as

$$f_{IF}(t) = \frac{1}{2\pi} \frac{d\varphi(t)}{dt}$$

where $\varphi(t)$ is the phase of the signal.

Because of its importance in practical, such as sonar, radar, speech and so on, the estimation of the IF have been studied extensively for many years. The coverage includes phase difference of the analytic signal, the recursive least squares and so on [2–5], these methods provide a good basis for us to understand the complicated problems of the IF estimation.

2.2. The linear canonical transform (LCT)

The LCT with parameter $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of a signal $x(t)$ is defined as follows [10]:

$$F^{\mathbf{A}}[x](u) = \begin{cases} \int_{-\infty}^{\infty} x(t) \sqrt{\frac{1}{j2\pi b}} e^{j(2)((a/b)t^2 - (2/b)ut + (d/b)u^2)} dt, & b \neq 0 \\ \sqrt{d} e^{j(2)c} \int_{-\infty}^{\infty} x(du) du, & b = 0 \end{cases} \quad (1)$$

where a, b, c, d are real numbers satisfying $ad - bc = 1$, i.e., $\det(\mathbf{A}) = 1$.

The inverse transform of the LCT (ILCT) is given by the LCT having parameter $\mathbf{A}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Hence, we can obtain the original signal $x(t)$ from $F^A[x](u)$ via

$$x(t) = F^{A^{-1}}[F^A[x](u)](t) = \begin{cases} \int_{-\infty}^{\infty} F^A[x](u) \sqrt{\frac{j}{2\pi b}} e^{(j/2)(-(d/b)u^2 + (2/b)ut - (a/b)t^2)} du, & b \neq 0 \\ \sqrt{a} e^{-(j/2)cat^2} F^A[x](at), & b = 0 \end{cases} \quad (2)$$

From Eq. (1), we can see that, when $b=0$, the LCT of a signal is essentially a chirp multiplication and it is of no particular interest to our object. Therefore, without loss of generality, we set $b>0$ in the following sections of the paper.

If we choose the parameter $\mathbf{A} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, the LCT will be reduced to the fractional Fourier transform (FRFT) and to Fourier transform (FT) when $\theta = \pi/2$. When the parameter $\mathbf{A} = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$, the LCT reduces to Fresnel transform. Besides, the LCT satisfies the additivity property as shown in [18].

With the achievements of the LCT in signal processing community, there emerges a large amount of new and interesting results associated with the LCT. The classical Shannon sampling theory has been generalized to the LCT domain [19,20], the well known convolution and product theorem [21,22], the uncertainty principles [23] and the spectral analysis [12] are well investigated in the LCT domain. Recently, more results are derived and proposed associate with the ambiguity function [24], the Wigner–Ville distribution [25] and the extrapolation method [26] in the linear canonical transform domain. For the best of our knowledge, there are no paper published associated with the IF estimation in the LCT domain, it is therefore interesting and worthwhile to study the IF estimation methods based on the LCT.

3. Instantaneous frequency estimation based on the LCT

Similarly with the power spectrum associated with the FT, the power spectrum associated with the LCT can be defined easily. Suppose $F^A[x](u)$ denotes the LCT of a signal $x(t)$, then the LCT power spectrum of this signal $x(t)$ can be defined as $|F^A[x](u)|^2$. Based on this definition, we can obtain the main results of this paper as the following theorem

Theorem 1. *Let a signal $x(t) = B(t)e^{j\varphi(t)}$, and $\varphi(t)$ is the phase of this signal. Then the instantaneous frequency of $x(t)$ associated with the LCT is equal to*

$$f_{IF}(t) = \frac{1}{2\pi} \frac{d\varphi(t)}{dt} = \frac{1}{2\pi} \left(\frac{M-N}{2|x(t)|^2 tb(a+d)} - \frac{a-d}{2b} t \right) \quad (3)$$

where

$$M = |F^{A^{-1}}[F^A(x(t))(u)u](t)|^2, \quad N = |F^A[F^{A^{-1}}(x(t))(u)u](t)|^2$$

Proof. By the properties of the LCT [18],

$$F^A[x(t)t](u) = (ud + jbD)F^A[x(t)](u) \quad (4)$$

where $D = d/dt$ is a differential operator.

So

$$F^{A^{-1}}[F^A(x(t))(u)u](t) = (ta - jbD)x(t) \quad (5)$$

and because

$$F^A[F^{A^{-1}}(x(t))](t) = x(t)$$

Based on this relationship, Eq. (5) can be rewritten as

$$F^A[F^{A^{-1}}(x(t))(u)u](t) = (td + jbD)x(t) \quad (6)$$

from Eqs. (5) and (6), we obtain

$$M = |F^{A^{-1}}[F^A(x(t))(u)](t)|^2 = |B(t)|^2 \left(ta + b \frac{d\varphi(t)}{dt} \right)^2 + b^2 \left(\frac{dB(t)}{dt} \right)^2$$

$$N = |F^A[F^{A^{-1}}(x(t))(u)u](t)|^2 = |B(t)|^2 \left(td - b \frac{d\varphi(t)}{dt} \right)^2 + b^2 \left(\frac{dB(t)}{dt} \right)^2$$

so

$$|B(t)|^2 t^2 (a^2 - d^2) + 2|B(t)|^2 \frac{d\varphi(t)}{dt} tb(a + d) = M - N$$

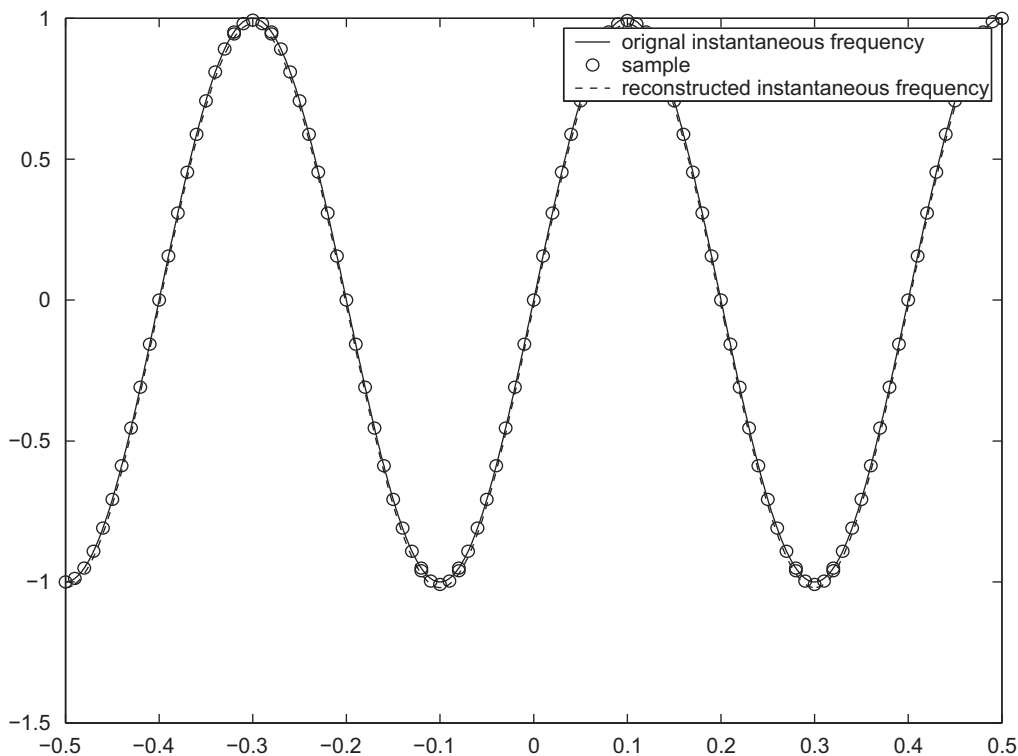


Fig. 1. The original and estimated IF.

$$\frac{d\varphi(t)}{dt} = \frac{M-N-|B(t)|^2 t^2 (a^2-d^2)}{2|B(t)|^2 tb(a+d)} = \frac{M-N}{2|B(t)|^2 tb(a+d)} - \frac{a-d}{2b} t$$

and notice,

$$|x(t)|^2 = |B(t)|^2$$

the final result can be obtained as

$$\frac{d\varphi(t)}{dt} = \frac{M-N}{2|x(t)|^2 tb(a+d)} - \frac{a-d}{2b} t \quad \square \quad (7)$$

Theorem 1 reflects the relationship between the IF of a signal and the LCT spectrum of this signal, in real application, we must use the discrete form of Eq. (3). Suppose the samples of the signal satisfies the conditions of the sampling theorem associated with the LCT [19,20]. We rewritten **Theorem 1** as the following corollary in discrete form.

Corollary 1. *If a signal is uniformly sampled, and the sampling interval is T . Let $\hat{f}_{IF}(t)$ denotes the estimated IF of this signal, then the discrete form of IF derived in Theorem 1 can*

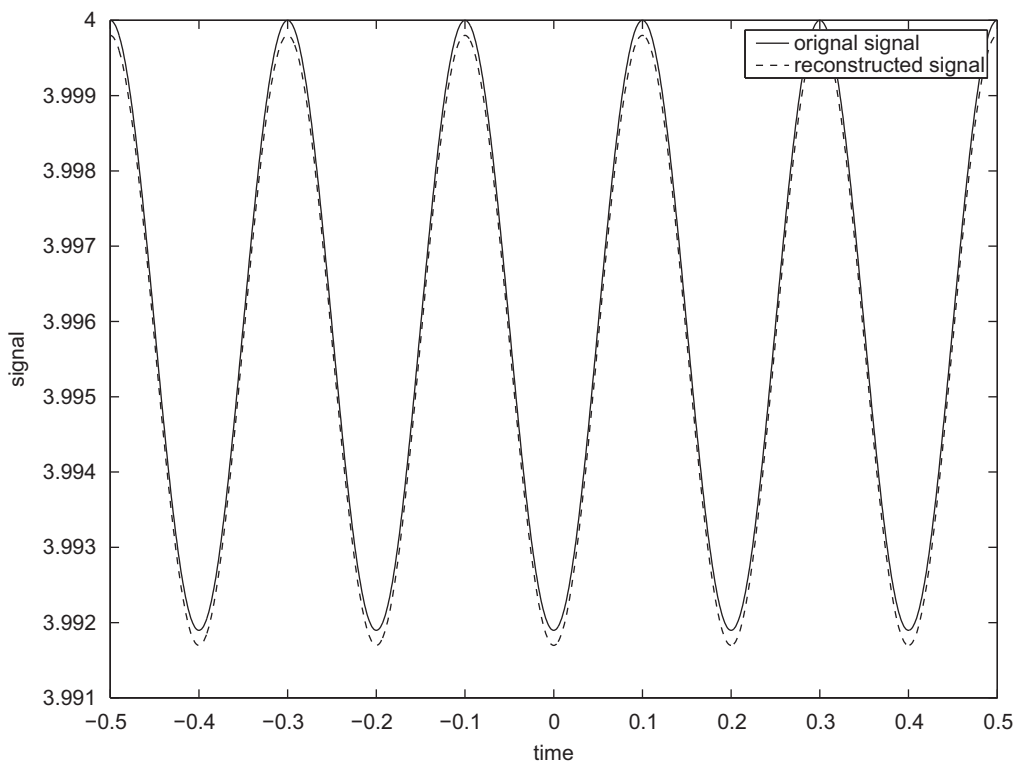


Fig. 2. The original and reconstructed signal.

be written as

$$\hat{f}(nT) = \frac{|F^{A^{-1}}[F^A(x(nT))(u)u](nT)|^2 - |F^A[F^{A^{-1}}(x(nT))(u)u](nT)|^2}{2|x(nT)|^2(nT)b(a+d)} - \frac{a-d}{2b}(nT) \quad (8)$$

Theorem 1 and **Corollary 1** can be looked as the IF estimation method associated with the LCT, from the relationship between the LCT and the fractional Fourier transform, these results can also be obtained in the fractional Fourier transform domain by choosing the different parameter A , we obtain the following corollary as the special cases of **Theorem 1**.

Corollary 2. When we choose the parameter $\mathbf{A} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, Eq. (3) will reduce to the IF associated with the FRFT, and to the IF associated with the Fourier transform with $\theta = \pi/2$.

4. Simulations

In order to verify the correctness of the derived results, two kinds of signals (with or without noise) will be used in this section to verify the derived results. From **Corollary 1**,

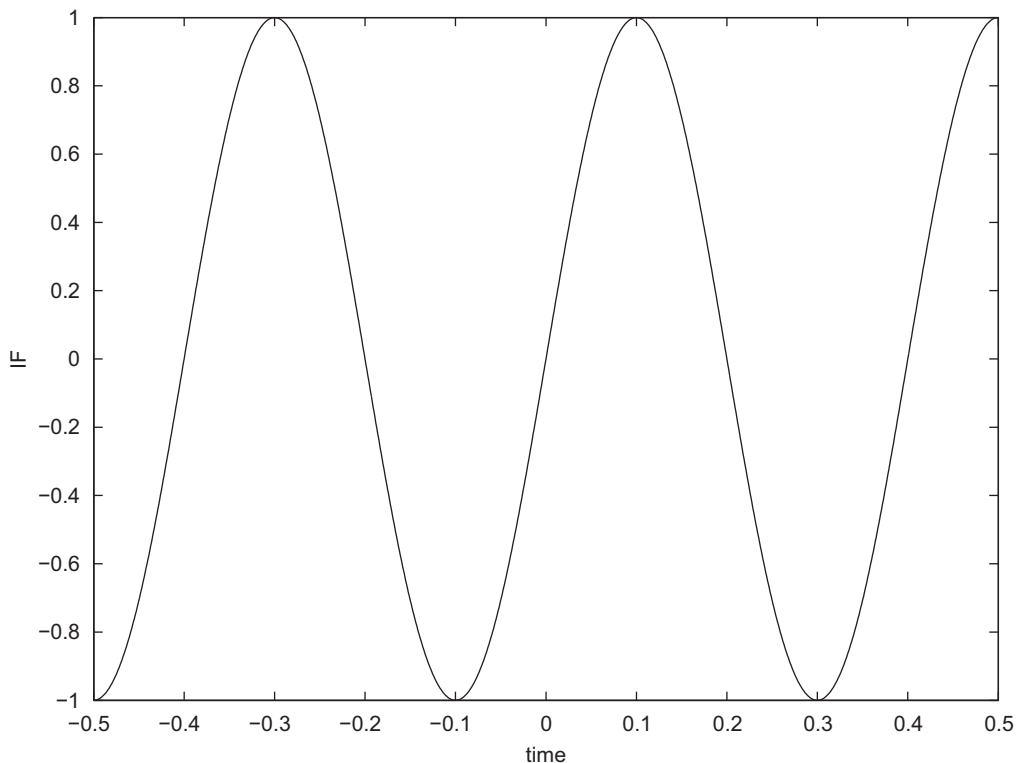


Fig. 3. The original IF.

when the IF is estimated, the reconstructed signal can be expressed as

$$\hat{x}(nT) = |x(nT)| \exp \left[j \sum_{n=-N}^N \hat{f}_{IF}(nT) T \right] \quad (9)$$

4.1. Signal without noise

In this subsection, the signal is chosen to be

$$x(t) = 4 \exp \left(j \int_{-1/2}^t \sin(5\pi t) dt \right), \quad t \in [-0.5, 0.5] \quad (10)$$

The number of sampling points is 100, and the sampling frequency is 100 Hz.

Fig. 1 plots the original IF $f_{IF}(nT)$ and the estimated IF $\hat{f}_{IF}(nT)$, respectively. Fig. 2 plots the original signal and the reconstructed signal. From these figures, we can see that this algorithm is not only effective but also has a high accuracy.

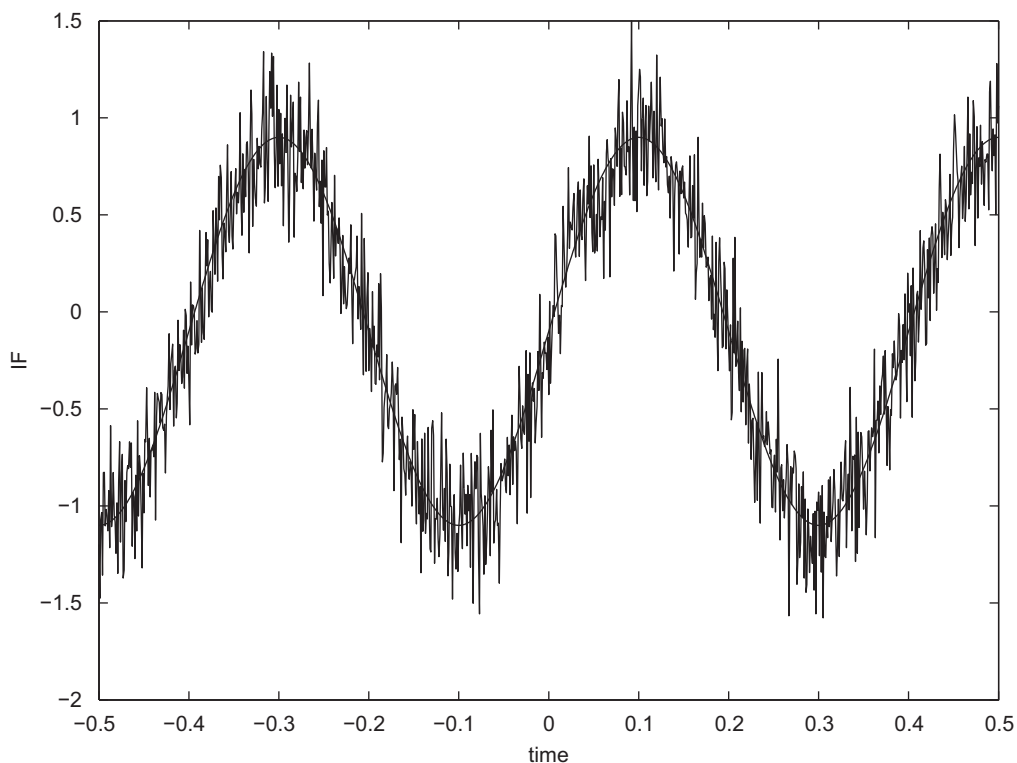


Fig. 4. The estimated IF when SNR=10 dB.

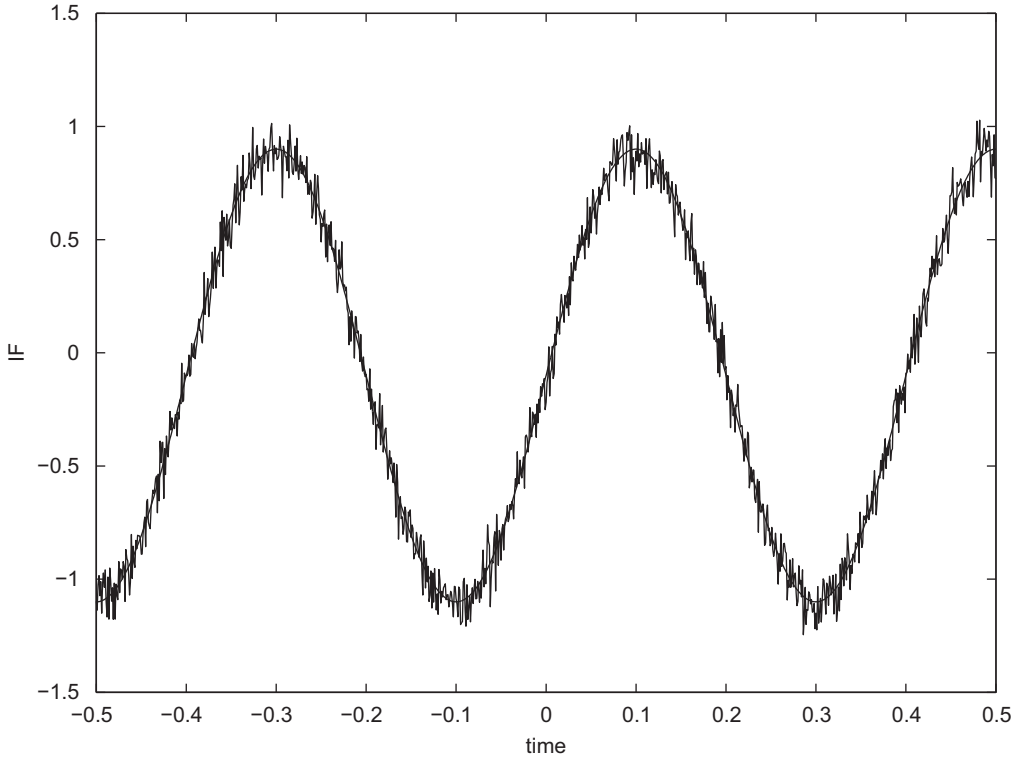


Fig. 5. The estimated IF when SNR=20 dB.

4.2. The signal with noise

When the signal in Section 4.1 is added with the Gauss white noise, so it can be rewritten as

$$x(t) = 4 \exp \left(j \int_{-1/2}^t \sin(5\pi t) dt + v(t) \right), \quad t \in [-0.5, 0.5] \quad (11)$$

where $v(t)$ is the Gauss white noise. The number of sampling points is 256, and the sampling frequency is 100 Hz, and Fig. 3 is the original IF of the signal $x(t)$.

Figs. 4 and 5 plot the original IF with SNR=10 dB and SNR=20 dB, respectively. Because of the noise, the estimation accuracy will decline. Although the lower of the signal-to-noise (SNR), the lower quality of estimation result, we can still estimate the IF as shown in this simulation.

5. Conclusion

This paper proposes a novel method of IF estimation based on the linear canonical transform. The simulation results show the effectiveness of above algorithms. The advantage of this method is non-iterative, and it does not need projection or peak-searching. Furthermore, it has a high capacity of anti-noise and a high estimation accuracy.

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